

Thanh Dang

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EDUCATION

California State Polytechnic University, Pomona

Bachelor of Science, Computer Science

May 2026

GPA: 3.73 / 4.00

- **Honors:** Magna Cum Laude, STEM Advantage Scholar

WORK EXPERIENCE

CPP Enterprises

Pomona, CA

AI/ML Researcher

May 2025 – Present

- Spearheaded multimodal AI pipeline processing 50+ hours of mobility scooter driving footage, using WhisperX speaker diarization and Google Gemma LLM to identify driving tasks and generate structured annotations
- Trained multi-task Transformer on pose keypoint sequences for bilateral asymmetry analysis and weak-side classification, achieving 91% rank accuracy for stroke-control separation and 92% AUC for weak-side prediction

Full Stack Developer

January 2026 – May 2026

- Developed GPU-accelerated video worker using Ray and Kafka to execute 3 asynchronous pipelines for pose estimation, task identification, and stability classification with fault-tolerant distributed processing
- Implemented NestJS REST APIs with PostgreSQL persistence for multi-tenant clinical workflows; streamed real-time video processing status via SSE and Redis pub/sub; deployed platform via Docker Compose on Jetstream2
- Led alignment meetings with 6-person dev team and collaborators to clarify scope and define deliverables

Cal Poly Pomona STARS Program

Pomona, CA

Undergraduate Research Assistant

October 2024 – May 2025

- Built Python desktop application for Delta NCSA HPC cluster file management via SFTP, streamlining upload, download, and deletion workflows; presented at Cal Poly Pomona RSCA Conference to 20+ attendees
- Utilized YOLO11 pose estimation with IoU-based person tracking to extract upper-body keypoints from videos
- Reduced dataset size of 40 mobility scooter driving sessions by 15% via temporal filtering of non-driving segments

theCoderSchool

Cerritos, CA

Code Coach

January 2024 – February 2025

- Led coding classes for 30+ elementary students, teaching programming through game development in Scratch
- Coached 5 students one-on-one in web development, teaching HTML, CSS, and JavaScript fundamentals
- Designed lesson plans and assigned homework to reinforce structured learning and programming concepts

CPP Software Engineer Association

Pomona, CA

Frontend Developer

August 2023 - May 2024

- Developed mobile platform to connect school organizations with members using React Native and Express.js
- Utilized JavaScript, CSS, and React Navigation to implement responsive UI and smooth screen transitions
- Collaborated in Agile team environment, participating in sprint planning and stand-ups to coordinate development

PROJECTS

TrailTalk

Backend Developer

September 2025 – December 2025

- Created social media web app for park enthusiasts using React and Firebase, featuring 63 U.S. national parks
- Designed scalable NoSQL data model supporting user accounts, messaging, and content management
- Implemented live private messaging with TypeScript and Firestore, including notifications and read receipts

Carbon Closet

Cal Poly Pomona Hackathon/Bronco hacks

Full Stack Developer

April 2025

- Built e-commerce platform for secondhand clothing using Next.js and Firebase; won 3rd place among 60 teams
- Developed interface with JavaScript and Tailwind CSS for live listings, searching, and gacha bundle purchases
- Containerized application with Docker and deployed to AWS EC2, gaining hands-on experience with cloud hosting

Recyclable or Not Bot

UC Riverside Hackathon/Rosehack

Backend Developer

January 2025

- Created computer vision web app to detect recyclability; awarded best sustainability track among 50+ teams
- Implemented object detection and inference using OpenCV frame capture and YOLOv8 deep learning model
- Integrated pipeline with Flask API, delivering real-time classification and color-coded score to React frontend

SKILLS

- **Programming Languages:** Python, Java, JavaScript, TypeScript, SQL, HTML, CSS
- **Frameworks:** React, React Native, Next.js, NestJS, Flask, PyTorch, Ray
- **Tools & Technologies:** Firebase, AWS, PostgreSQL, MySQL, Redis, Docker, Kafka, Node.js, Git